Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

ISWE406P – Agile Development Process and Devops Lab

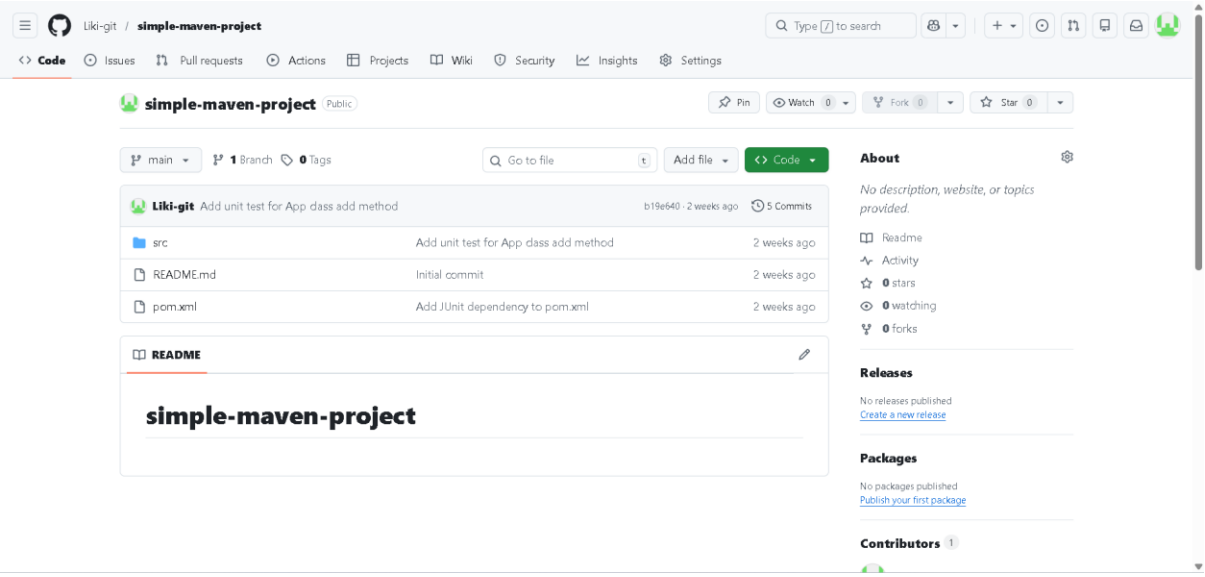
Assignment III

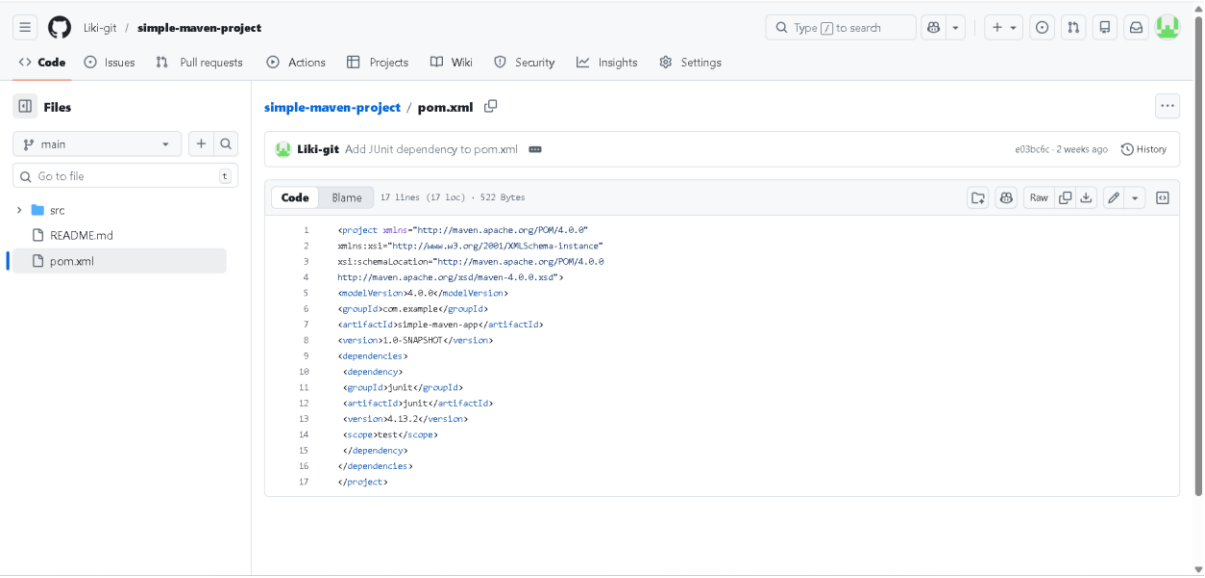
NAME : P.LIKITHA

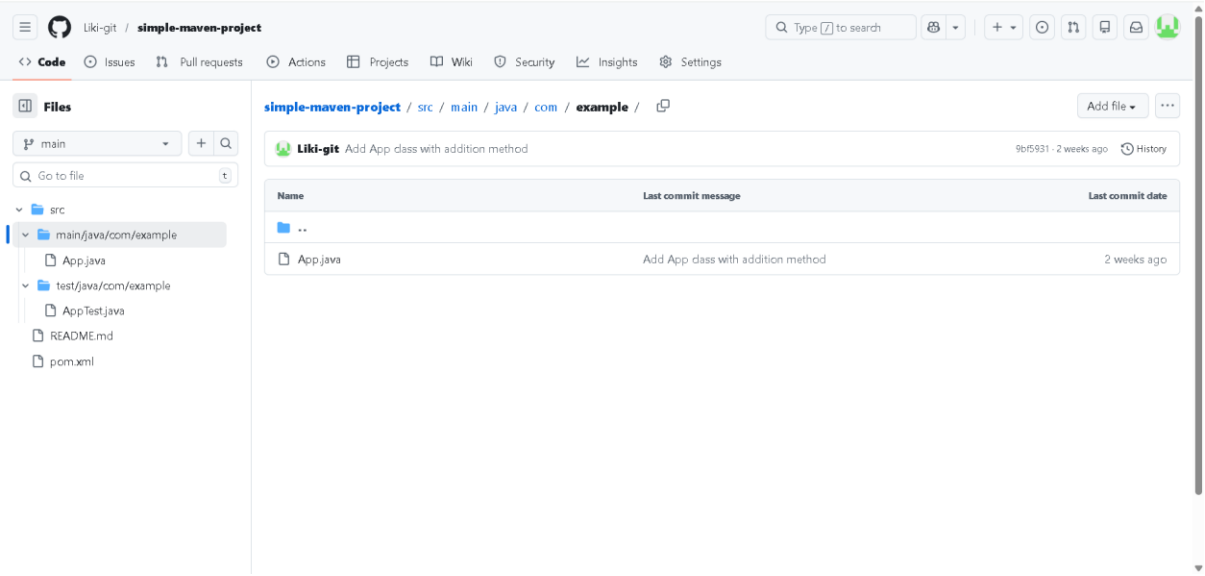
REG NO. : 23MIS0243

SLOT : L49+L50

1. Create a Maven Project in Git

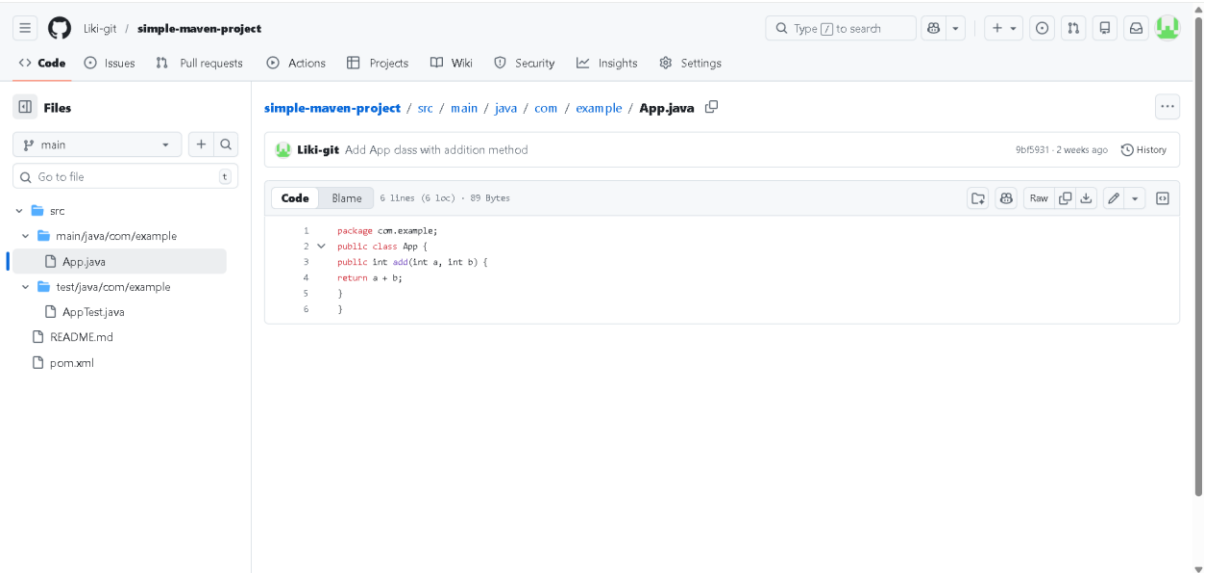


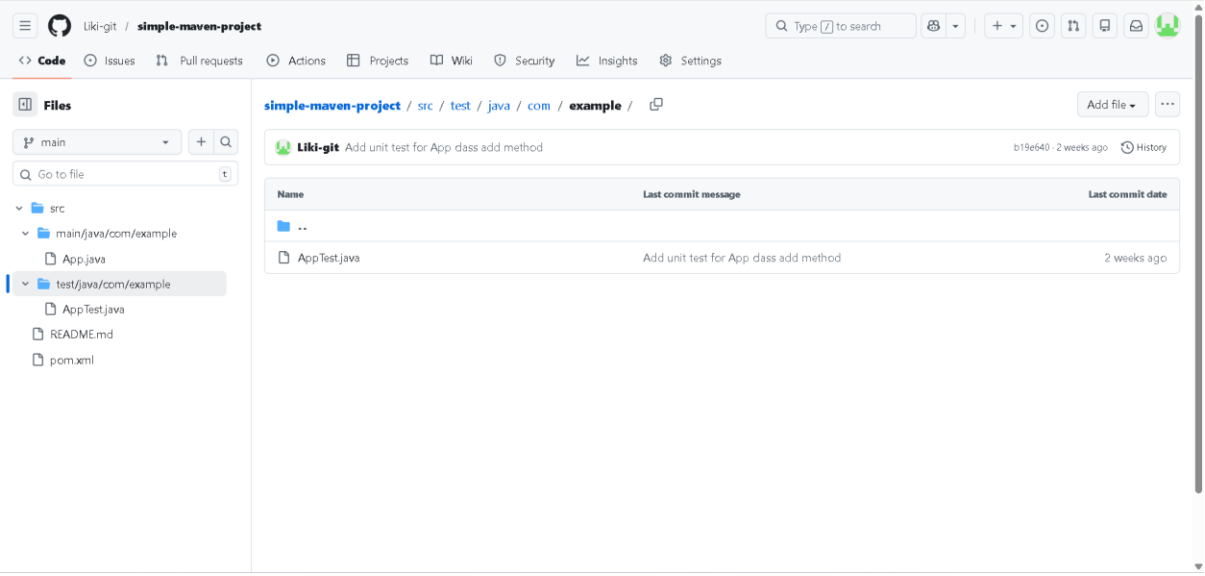


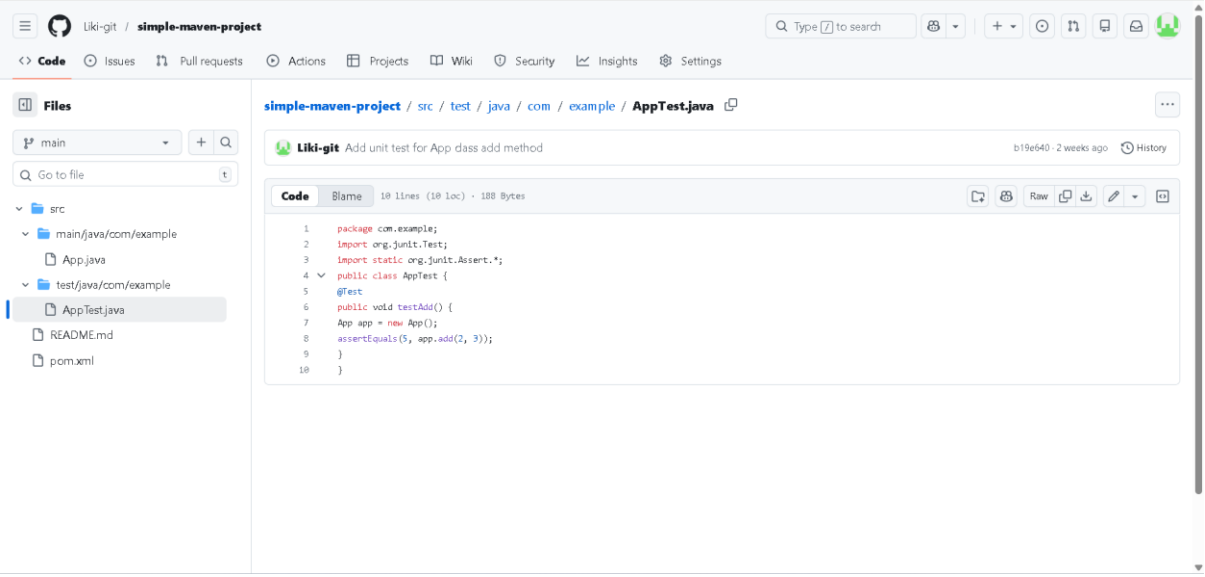


2. Implement the Java Program

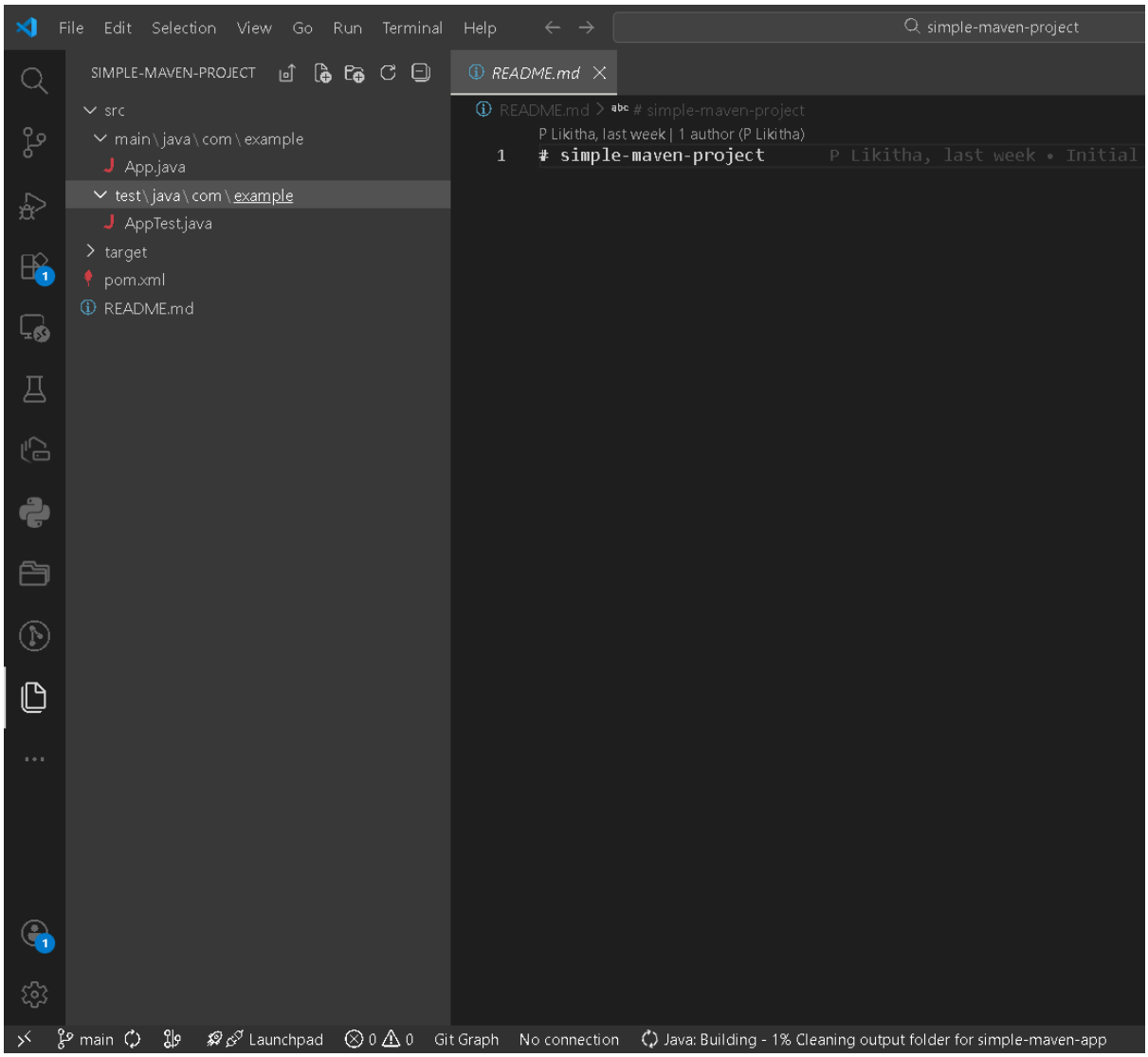
Simple Calculator Application







3. Push and Clone the Repository (Vs Code)



4. Build the Maven Project

Compile the project

```
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> mvn -v
Apache Maven 3.9.12 (848fbb4bf2d427b72bdb2471c22fced7ebd9ggram Fgram Files\Java\jdk-21.0.10
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:simple-maven-app >-------------------
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ simple-maven-app ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.991 s
[INFO] Finished at: 2026-03-03T12:03:56+05:30
[INFO] ------------------------------------------------------------------------
```

Run tests

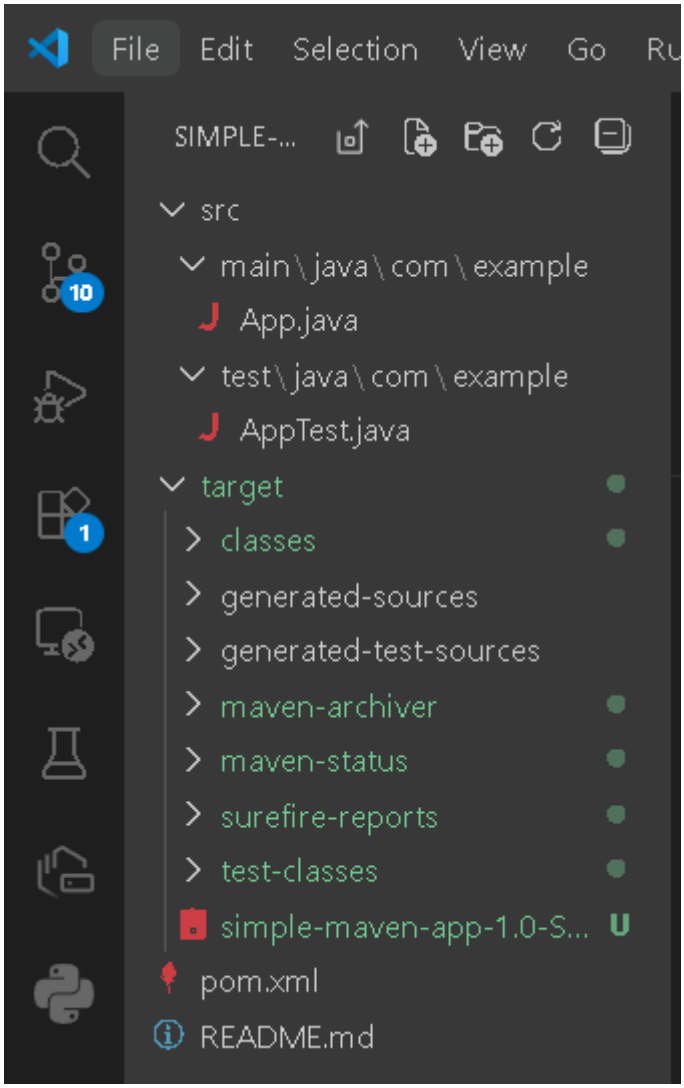
```
generated-sources
generated-test-sources
maven-status
surefire-reports
test-classes
pom.xml
README.md

[INFO] ------------------------------------------------------------------------
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:simple-maven-app >-------------------
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ simple-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ simple-maven-app ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ simple-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.2.5/surefire-junit4-3.2.5.pom
```

```
generated-sources
generated-test-sources
maven-status
surefire-reports
test-classes
pom.xml
README.md

[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ simple-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.2.5/surefire-junit4-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.2.5/surefire-junit4-3.2.5.pom (3.0 kB at 2.6 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/3.2.5/surefire-providers-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/3.2.5/surefire-providers-3.2.5.pom (2.6 kB at 4.4 kB/
s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.2.5/common-junit4-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.2.5/common-junit4-3.2.5.pom (3.1 kB at 8.9 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.2.5/common-junit3-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.2.5/common-junit3-3.2.5.pom (2.8 kB at 14 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.2.5/common-java5-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.2.5/common-java5-3.2.5.pom (2.8 kB at 16 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.2.5/common-junit4-3.2.5.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.2.5/surefire-junit4-3.2.5.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.2.5/common-junit3-3.2.5.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.2.5/common-java5-3.2.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.2.5/surefire-junit4-3.2.5.jar (18 kB at 100 kB/s)
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.112 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
main*  Launchpad  0 2  Git Graph  Indexing completed.  Java: Ready  Finish Setup  Go Live
```

```
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.112 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  7.189 s
[INFO] Finished at: 2026-03-03T12:04:19+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project>
```

Build the project (mvn clean install)

Create .jar file inside: target/simple-maven-app-1.0-SNAPSHOT.jar



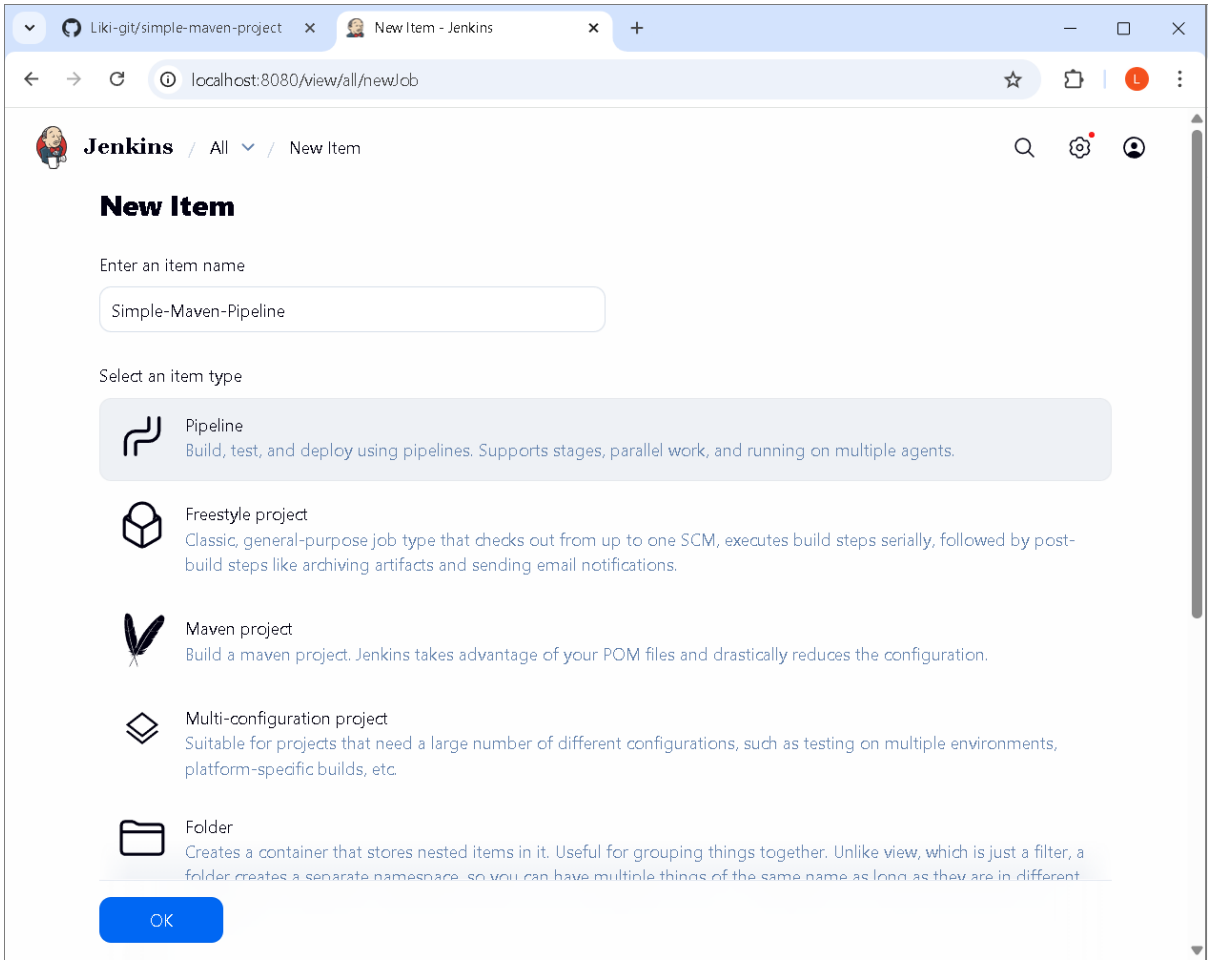
```
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:simple-maven-app >-------------------
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.2/maven-install-plugin-3.1.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.2/maven-install-plugin-3.1.2.pom (8.5 kB a
t 9.1 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ simple-maven-app ---
[INFO] Deleting C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ simple-maven-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\src\test\resources
```

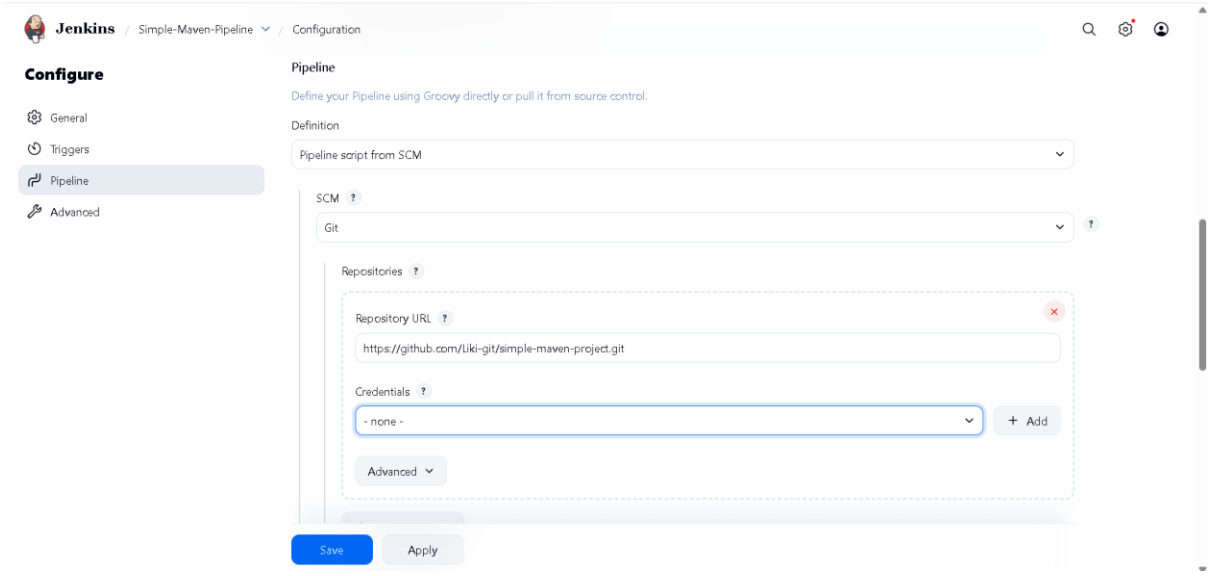
```
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ simple-maven-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ simple-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.105 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO]
[INFO]
```

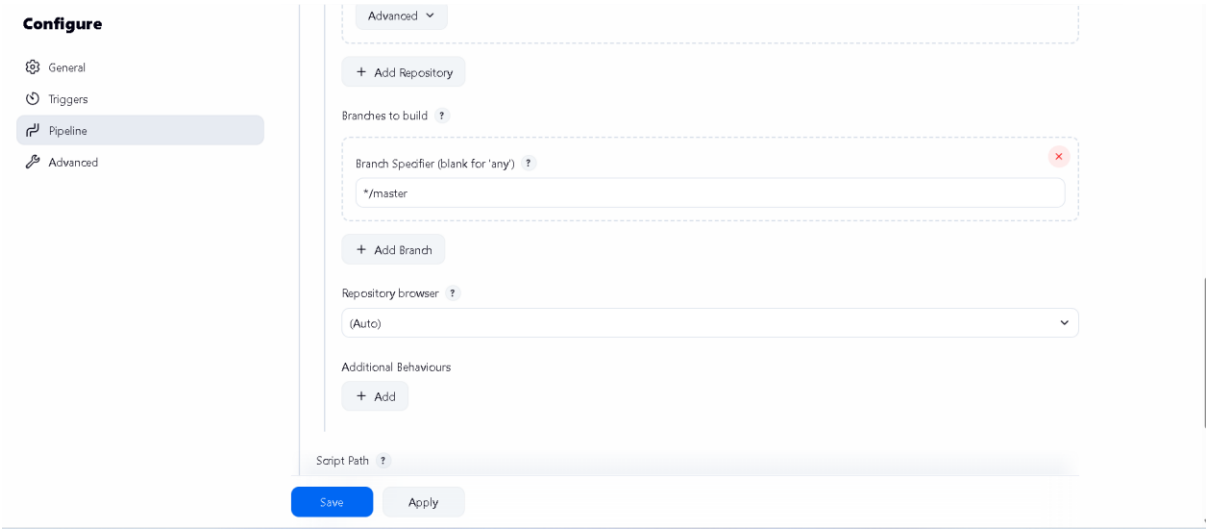
```
[INFO] --- jar:3.4.1:jar (default-jar) @ simple-maven-app ---
[INFO] Building jar: C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\target\simple-maven-app-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ simple-maven-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.pom (2.9 kB
at 18 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.18/maven-resolver-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.18/maven-resolver-1.9.18.pom (22 kB at 111 kB/s
)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.pom (2.7 kB at
 14 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar (157 kB at
 320 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB
at 244 kB/s)
[INFO] Installing C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\pom.xml to C:\Users\likit\.m2\repository\com\example\simple-maven-app\
1.0-SNAPSHOT\simple-maven-app-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project\target\simple-maven-app-1.0-SNAPSHOT.jar to C:\Users\likit\.m2\reposit
ory\com\example\simple-maven-app\1.0-SNAPSHOT\simple-maven-app-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  7.291 s
[INFO] Finished at: 2026-03-03T12:10:51+05:30
[INFO] ------------------------------------------------------------------------
```

5. Create and Run Jenkins Pipeline

Create a new Pipeline project in Jenkins.







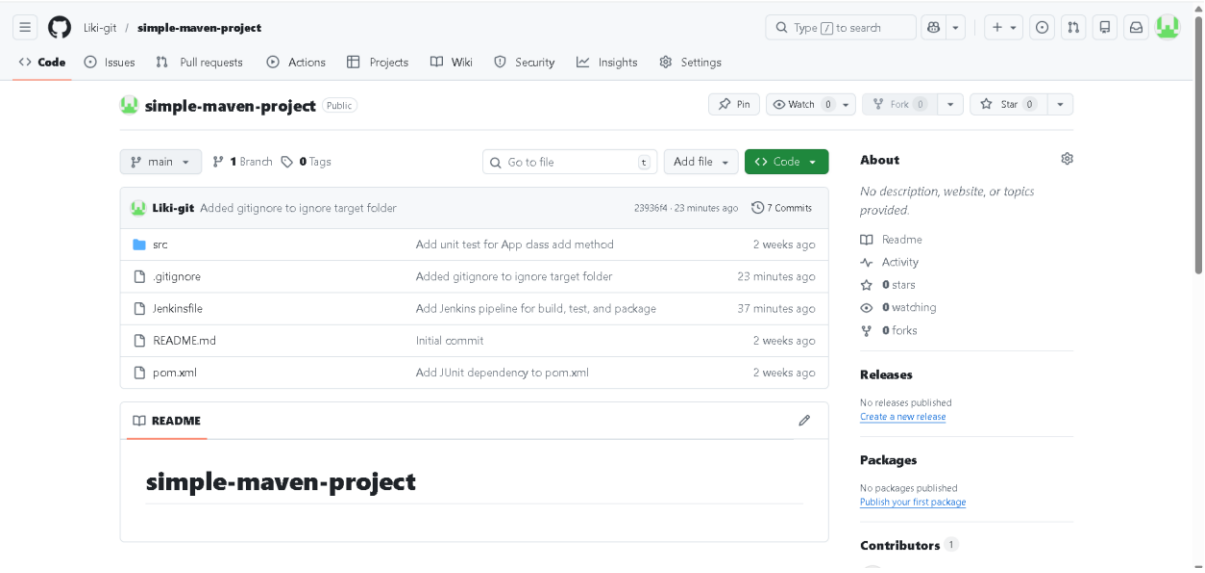

```
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> git pull origin main
From https://github.com/Liki-git/simple-maven-project
 * branch            main       -> FETCH_HEAD
Updating b19e640..5682c8e
Fast-forward
 Jenkinsfile | 37 ++++++++++++++++++++++++++++++++++++
 1 file changed, 37 insertions(+)
 create mode 100644 Jenkinsfile
```

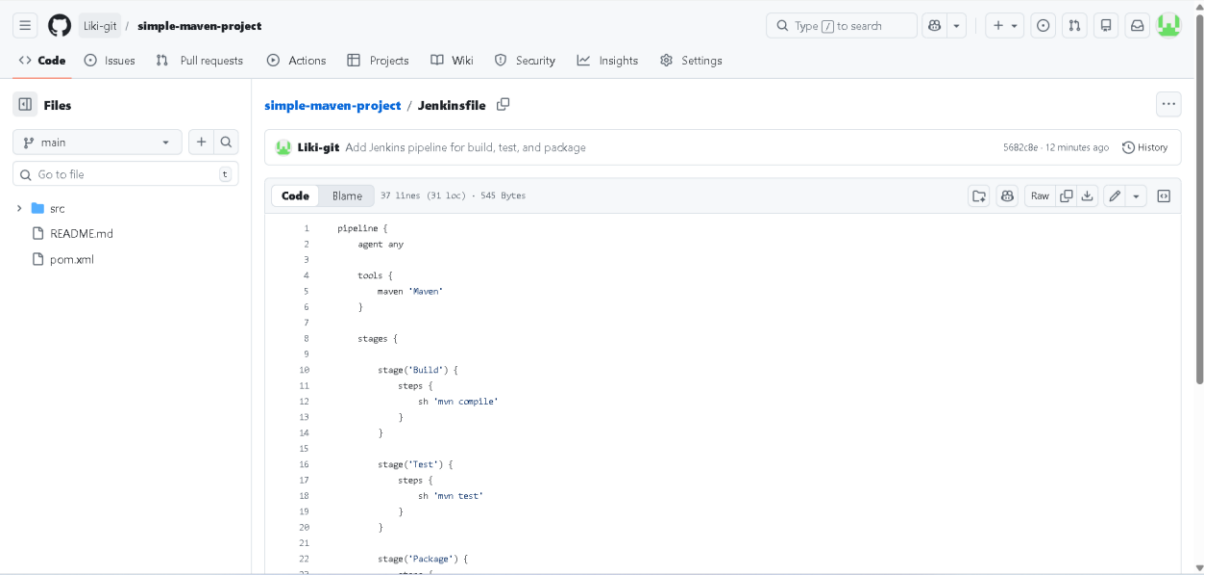
```
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> ni .gitignore


    Directory: C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        03-03-2026     12:32              0 .gitignore


PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> notepad .gitignore
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project>
```

Create a pipeline script

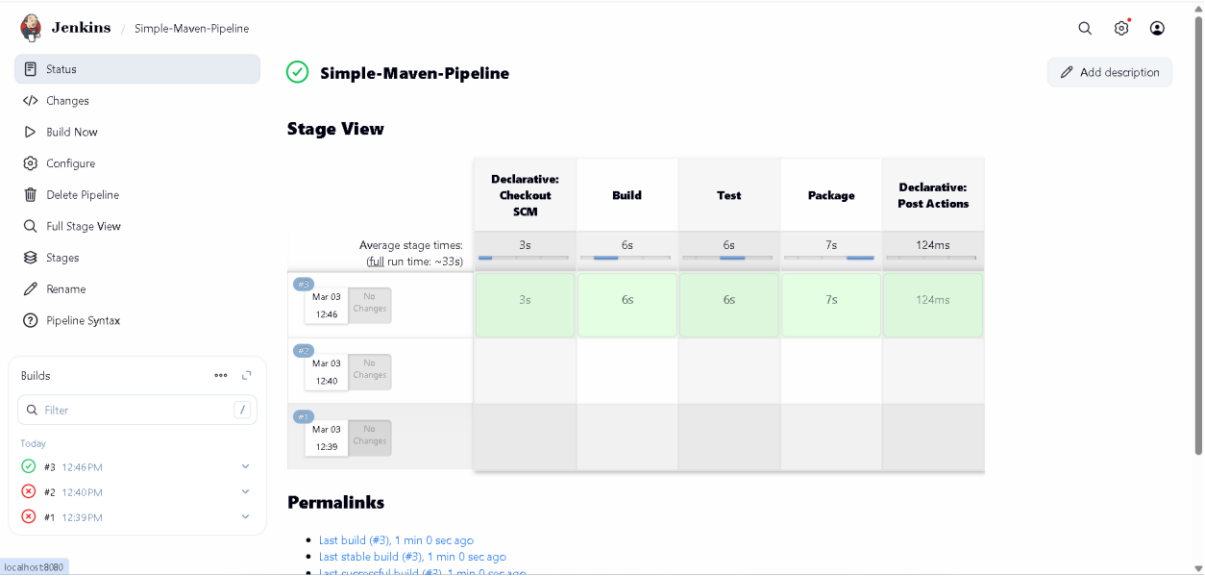
```
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> notepad .gitignore
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> git add .gitignore
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> git commit -m "Added gitignore to ignore target folder"
[main 23936f4] Added gitignore to ignore target folder
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 294 bytes | 98.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Liki-git/simple-maven-project.git
   5682c8e..23936f4  main -> main
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project>
```

```
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\likit\OneDrive\Desktop\Agile_asses1\simple-maven-project>
```





Execute the pipeline and verify successful build completion.



Jenkins / Simple-Maven-Pipeline / #3 / Console Output

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#3'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

Console Output

Download | Copy | View as plain text

```
Started by user P Likitha
Obtained Jenkinsfile from git https://github.com/Liki-git/simple-maven-project.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\likit\.jenkins\workspace\Simple-Maven-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/Liki-git/simple-maven-project.git
 > C:\Program Files\Git\bin\git.exe init C:\Users\likit\.jenkins\workspace\Simple-Maven-Pipeline # timeout=10
Fetching upstream changes from https://github.com/Liki-git/simple-maven-project.git
 > C:\Program Files\Git\bin\git.exe --version # timeout=10
 > git --version # 'git version 2.53.0.windows.1'
 > C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/Liki-git/simple-maven-project.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/Liki-git/simple-maven-project.git # timeout=10
 > C:\Program Files\Git\bin\git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 677701d0e5cc5f02ee6128ee0e3dfcfd5edec662 (refs/remotes/origin/main)
```

Jenkins / Simple-Maven-Pipeline / #3 / Console Output

```
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ simple-maven-app ---
[INFO] Building jar: C:\Users\likit\.jenkins\workspace\Simple-Maven-Pipeline\target\simple-maven-app-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.189 s
[INFO] Finished at: 2026-03-03T12:46:47+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Build Successful!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```